

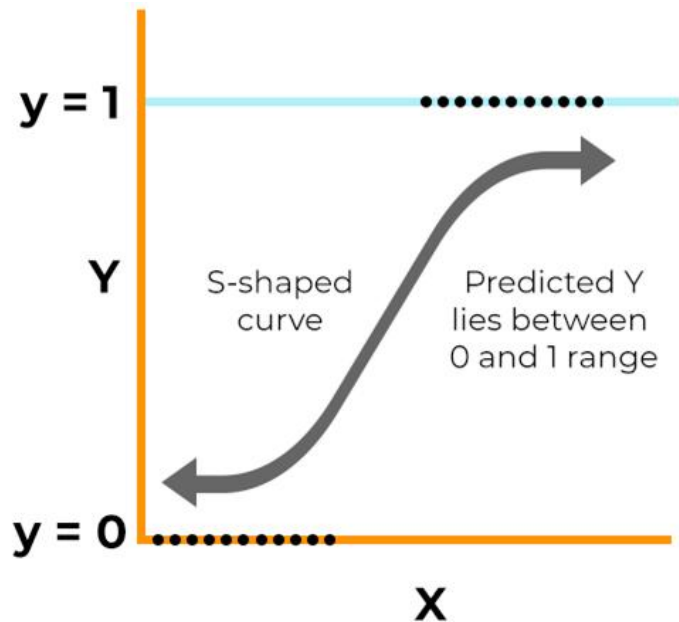
Agenda

Classification Techniques

Sci-kit Learn

Logistic Regression

Logistic regression is a **supervised** machine learning algorithm that accomplishes **binary classification** tasks by predicting the probability of an outcome, event, or observation.



Different Classification Techniques

Decision Tree Classifier

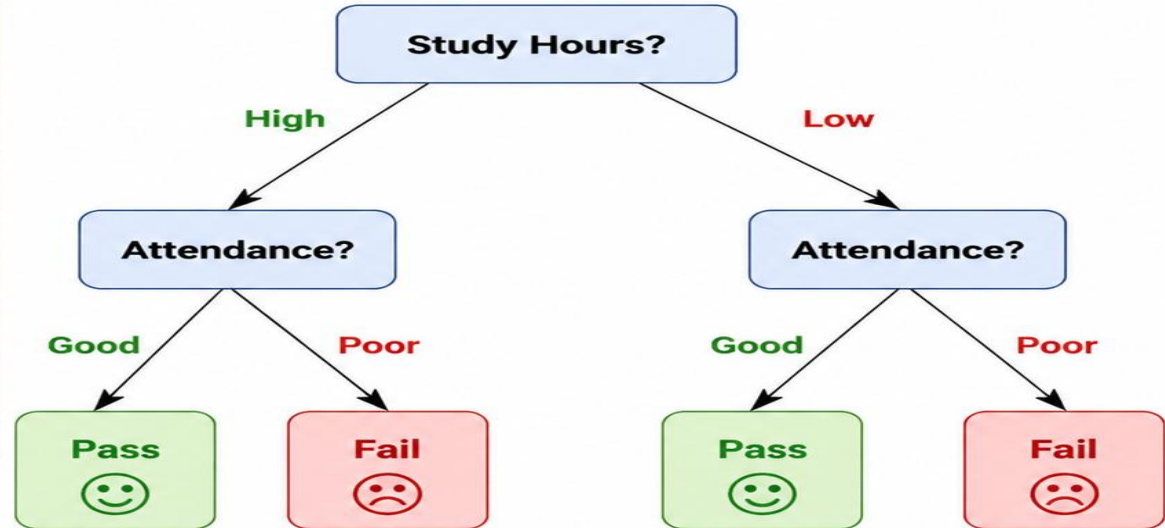
A decision tree classifier makes decisions by **splitting the data** into **smaller subsets** based on the values of input features, ultimately assigning a class label to each data point. It constructs a tree-like structure of decision rules that can be used for prediction and is easy to interpret.

Decision Tree Classifier – Simple Example

Goal: Predict whether a student will **Pass** or **Fail** based on **Study Hours** and **Attendance**.

Training Data

Study Hours	Attendance	Result
High	Good	Pass
High	Poor	Fail
Low	Good	Pass
Low	Poor	Fail
High	Good	Pass
Low	Good	Pass
High	Poor	Fail
Low	Poor	Fail



Example Prediction:

New Student → Study Hours = High, Attendance = Poor

Path: Study Hours (High) → Attendance (Poor) → Result = **Fail**

Decision Tree Classifier – Titanic Dataset Example

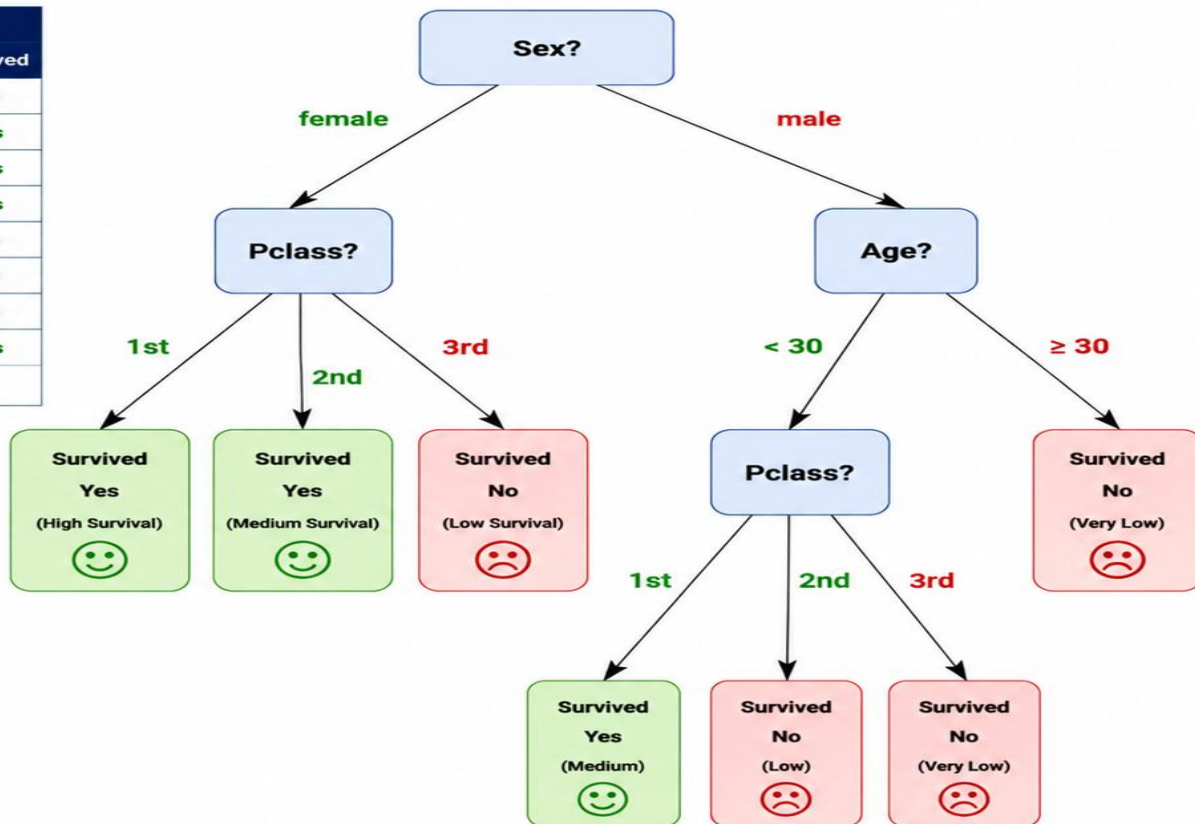
Goal: Predict whether a passenger **Survived** or **Did Not Survive** based on features like Sex, Pclass and Age.

Sample Titanic Data				
PassengerId	Pclass	Sex	Age	Survived
1	3	male	22	No
2	1	female	38	Yes
3	3	female	26	Yes
4	1	female	35	Yes
5	3	male	35	No
6	3	male	~	No
7	1	male	54	No
8	3	female	27	Yes
...	...	...	...	...

How to read the tree?

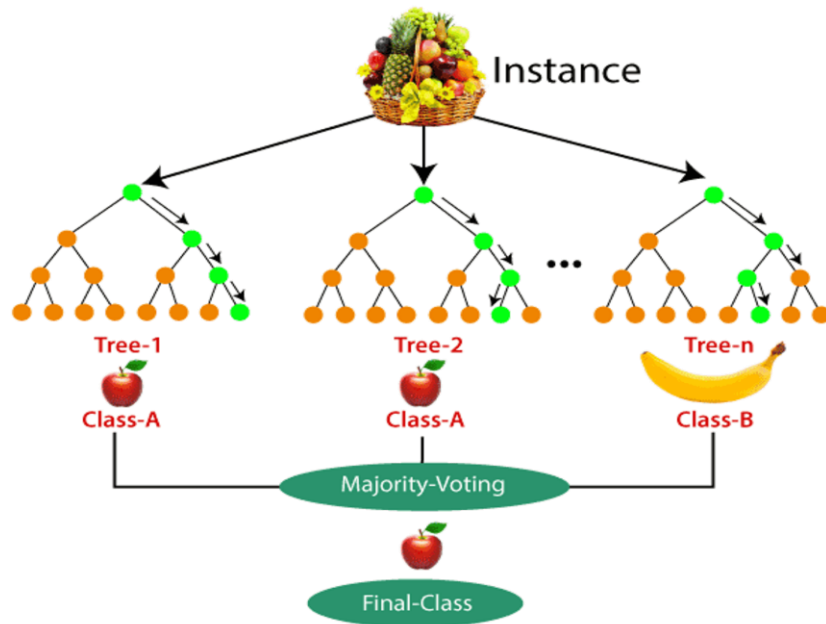
- Start from the top node.
- Follow the branch that matches the passenger's information.
- Reach the leaf node to get the prediction (Survived Yes/No).

-  Decision Node
-  Leaf Node (Survived = Yes)
-  Leaf Node (Survived = No)



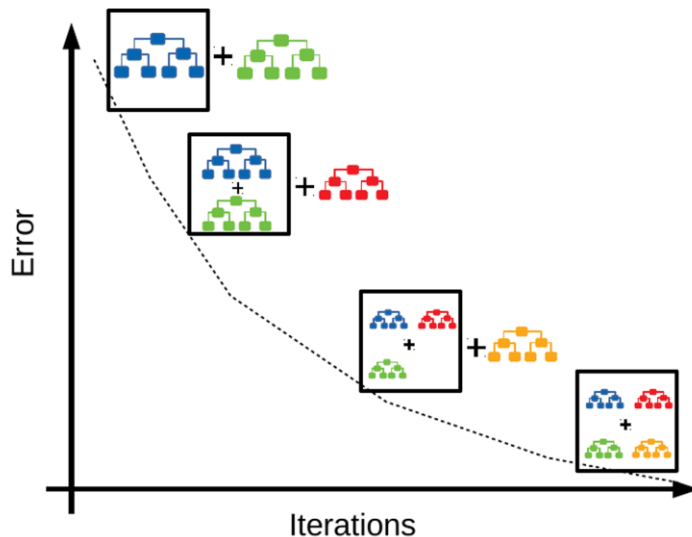
Random Forest Classifier

Random Forest is **ensemble** learning method that creates **multiple decision trees** during training and outputs the **average prediction** from all individual trees.



Gradient Boosting Classifier

Gradient Boosting Classifier is a **powerful ensemble learning technique** used for **classification tasks**. It belongs to the class of boosting algorithms, which build a strong predictive model by combining multiple weak learners sequentially.

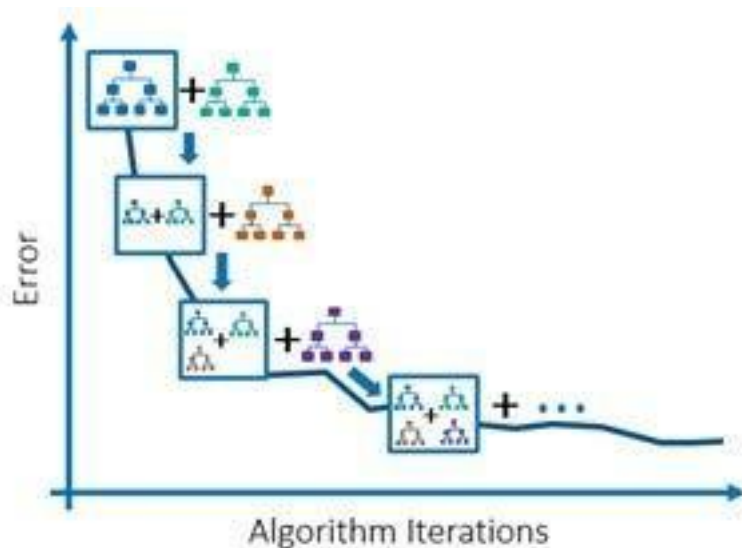


Ensemble:

Ensemble learning is a **machine learning** technique that **combines** the predictions of **multiple individual models** (learners) to **improve overall performance** and generalization. Instead of relying on a single model, ensemble methods create a "**committee**" of models and aggregate their predictions to make a final decision.

Gradient Boosting classifier

Gradient Boosting is **ensemble** learning technique that builds **multiple decision trees** sequentially, each one **correcting the errors of its predecessor**.



Accuracy

It is the ratio of number of **correct predictions** to the **total number of input samples**.

$$\text{Accuracy} = \frac{\text{Number of Correct predictions}}{\text{Total number of Predictions}}$$

Note:It works well only if there are **equal** number of samples belonging to **each class**.